MULTIVIEW: A Portable C99 System for Two-Camera Scene Analysis and 3D Model Extraction

Project documentation — v0.1.0

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Abstract

MULTIVIEW is a small, dependency-free C99 library for metric scene analysis and 3D model extraction from the synchronized video streams of *two stationary, calibrated cameras* observing an overlapping volume, optionally augmented by an unordered collection of still photographs and, in a later stage, range-finder measurements. This document is self-contained: it states what the system can do today (section 1), develops the underlying multiview-geometry theory from the camera model up (section 3–section 6), describes the numerical choices made in the implementation (section 9), and reports first quantitative results on a controlled synthetic scene (section 10). It is written to support later self-study and incremental development of the code base in a controlled environment: every claim in section 10 is reproducible from the deterministic test programs in the repository.

Contents

1 Introduction and capabilities

The target deployment is deliberately narrow and fully under our control: two cameras are rigidly mounted, their intrinsic and extrinsic calibration is known (or measured once, offline), and both observe the same working volume. Video from this rig is processed frame by frame; because the cameras do not move, all geometric quantities that depend only on the rig — projection matrices, the fundamental matrix, the rectifying homographies — are computed *once* and reused for every frame, which is the central efficiency win of the stationary-rig assumption.

The implemented core (v0.1.0) provides:

1. a calibrated pinhole camera model with Brown radial–tangential lens distortion, forward projection and ray back-projection (section 3);
2. exact two-view epipolar geometry from the calibration, and the normalized 8-point algorithm to estimate or verify it from image correspondences (section 4);
3. N-view linear triangulation of corresponding image points into metric 3D, with reprojection-error reporting (section 5);
4. planar stereo rectification of the pair, and dense correspondence by block matching with sub-pixel refinement, giving per-pixel depth (section 6);
5. point-cloud assembly and PLY export for inspection in standard mesh tools, plus minimal PGM image I/O (section 7).

Planned extensions — structure-from-motion over an auxiliary photo collection and fusion of range-finder data — are designed but not yet implemented; their theory and integration points are described in section 8 so that the code can grow into them without rearchitecting.

Design constraints. The library is strict C99 with no dependencies beyond `libm`, row-major `double` arrays throughout, and fully deterministic (no clocks, no ambient randomness; the demo uses a fixed-seed generator). This keeps it portable from desktop to embedded targets, and keeps every experiment repeatable — a prerequisite for the controlled-environment, learning-oriented use the project is intended for.

2 System overview

The per-frame processing pipeline is:

$$\underbrace{\text{2 video frames}}_{\text{capture}} \to \underbrace{\text{undistort}}_{\texttt{cam}} \to \underbrace{\text{rectify}}_{\texttt{rectify}} \to \underbrace{\text{dense match}}_{\texttt{stereo}} \to \underbrace{\text{depth / triangulate}}_{\texttt{triangulate}} \to \underbrace{\text{cloud accumulation}}_{\texttt{cloud}}$$

Sparse operation (feature correspondences instead of dense matching) uses the same chain with `epipolar` providing the correspondence-validation gate. Because the rig is static, temporal accumulation over video frames is a simple union of per-frame clouds in the fixed world frame; moving objects appear as temporally inconsistent points and can be segmented by comparing frames, which is the entry point for scene *analysis* beyond reconstruction.

Module	Header	Responsibility
`mat`	`mv/mat.h`	small dense linear algebra, Jacobi SVD, null vectors
`cam`	`mv/cam.h`	camera model, projection, rays, distortion
`epipolar`	`mv/epipolar.h`	$\mathbf{F}$ from calibration; normalized 8-point; $\mathbf{E}$
`triangulate`	`mv/triangulate.h`	N-view DLT triangulation, reprojection RMS
`rectify`	`mv/rectify.h`	Fusiello rectification, disparity→depth
`stereo`	`mv/stereo.h`	dense block matching (SAD, sub-pixel)
`img`	`mv/img.h`	PGM (P5) image I/O
`cloud`	`mv/cloud.h`	point cloud container, PLY export

Table 1: Module layout. `mv/mv.h` is the umbrella header.

3 Camera model and calibration

A camera is modeled as a pinhole with intrinsic matrix

$$\mathbf{K} = \begin{pmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix}, \tag{1}$$

rotation $\mathbf{R}$ and translation $\mathbf{t}$ mapping world points into the camera frame, $\mathbf{X}_c = \mathbf{R}\mathbf{X} + \mathbf{t}$; the camera center is $\mathbf{C} = -\mathbf{R}^\top \mathbf{t}$. A world point projects to the homogeneous pixel

$$\mathbf{x} \;\simeq\; \mathbf{P}\begin{pmatrix}\mathbf{X}\\ 1\end{pmatrix}, \qquad \mathbf{P} = \mathbf{K}\,[\,\mathbf{R} \mid \mathbf{t}\,], \tag{2}$$

where $\simeq$ denotes equality up to scale [HZ04]. Real lenses deviate from (2); we use the Brown model on normalized coordinates (x, y), $r^2 = x^2+y^2$:

$$\begin{aligned} x_d &= x\,(1 + k_1 r^2 + k_2 r^4 + k_3 r^6) + 2p_1 xy + p_2(r^2 + 2x^2), \\ y_d &= y\,(1 + k_1 r^2 + k_2 r^4 + k_3 r^6) + p_1(r^2 + 2y^2) + 2p_2 xy. \end{aligned} \tag{3}$$

Back-projection inverts (3) by fixed-point iteration (20 iterations suffice for the moderate distortions of machine-vision lenses) and returns the ray $\mathbf{C} + s\,\mathbf{d}$, $s > 0$.

Calibration itself is treated as an *input*: the rig is calibrated once offline with standard planar-target methods [Zha00, Tsa87], and the resulting $(\mathbf{K}, \mathbf{R}, \mathbf{t}, k_{1..5})$ per camera is what section 4–section 6 consume. The stationary rig makes this a one-time cost and removes online self-calibration from the critical path.

4 Two-view epipolar geometry

Corresponding pixels of the two views are linked by the fundamental matrix $\mathbf{F}$ through the epipolar constraint

$$\mathbf{x}_2^\top \mathbf{F}\, \mathbf{x}_1 = 0, \tag{4}$$

with $\mathbf{F}$ of rank 2; the line $\boldsymbol{\ell}_2 = \mathbf{F}\mathbf{x}_1$ is the locus in image 2 of possible matches of $\mathbf{x}_1$ (Figure 1). For *calibrated* views the essential matrix $\mathbf{E} = \mathbf{K}_2^\top \mathbf{F}\, \mathbf{K}_1 = [\mathbf{t}]_\times \mathbf{R}$ carries the same constraint in normalized coordinates [LH81].

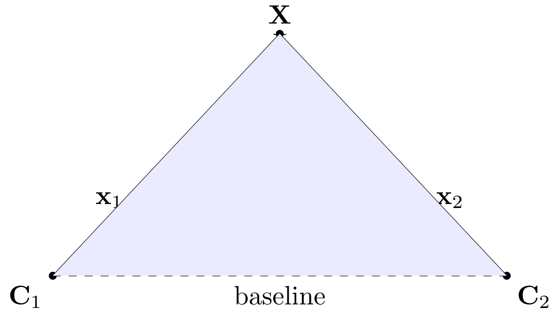

Figure 1: Epipolar geometry: the camera centers $\mathbf{C}_1, \mathbf{C}_2$ and the scene point $\mathbf{X}$ span the epipolar plane (shaded); its intersections with the image planes π_1, π_2 are the corresponding epipolar lines on which $\mathbf{x}_1$ and $\mathbf{x}_2$ must lie.

Because our rig is calibrated, $\mathbf{F}$ is available in closed form,

$$\mathbf{F} = [\, e_2 \,]_\times \, \mathbf{P}_2 \, \mathbf{P}_1^+, \qquad e_2 = \mathbf{P}_2 \begin{pmatrix} \mathbf{C}_1 \\ 1 \end{pmatrix}, \tag{5}$$

where $\mathbf{P}_1^+$ is the pseudo-inverse of $\mathbf{P}_1$ [HZ04]. The system computes (5) once at start-up and uses it for *correspondence gating*: any candidate match whose symmetric epipolar distance exceeds a threshold is rejected before triangulation.

Independently, $\mathbf{F}$ can be *estimated* from $n \geq 8$ point correspondences by the normalized 8-point algorithm [Har97]: each pair contributes one linear equation in the entries of $\mathbf{F}$, the points being first translated and scaled to centroid 0 and RMS radius $\sqrt{2}$ (normalization is essential for conditioning, see section 9); the solution is the smallest right singular vector of the design matrix, followed by rank-2 enforcement via SVD. In this system the estimator serves two purposes: (i) *health monitoring* — if the estimate from live correspondences drifts from the calibrated (5), the rig has been bumped or the calibration is stale; (ii) bootstrapping geometry for *uncalibrated* auxiliary photos (section 8). On contaminated correspondence sets the estimator is wrapped in a RANSAC loop [FB81] (planned with the feature layer).

5 Triangulation

Given observations $\mathbf{x}^{(v)} = (u_v, w_v)$ of the same point in $N \geq 2$ views with projection matrices $\mathbf{P}^{(v)}$, each view contributes two linear equations obtained by eliminating the projective scale from (2):

$$\begin{pmatrix} u_v \, \mathbf{P}^{(v)}_{3,:} - \mathbf{P}^{(v)}_{1,:} \\ w_v \, \mathbf{P}^{(v)}_{3,:} - \mathbf{P}^{(v)}_{2,:} \end{pmatrix} \begin{pmatrix} \mathbf{X} \\ 1 \end{pmatrix} = \mathbf{0}, \tag{6}$$

and stacking all views gives a $2N \times 4$ homogeneous system whose least-squares solution (smallest right singular vector, then dehomogenization) is the DLT triangulation [HZ04]. The implementation accepts any $N \geq 2$, so still photographs registered into the rig frame (section 8) simply extend the same call. DLT minimizes an algebraic residual; the gold-standard reprojection-optimal two-view method of Hartley–Sturm [HS97] is a planned refinement, worthwhile mainly for wide-baseline auxiliary views. The library reports the RMS reprojection error of every triangulated point, which is the quantity an eventual bundle adjustment would minimize [TMHF00].

6 Rectification and dense stereo

For dense per-pixel depth the pair is first *rectified*: both cameras are rotated (synthetically, about their centers) onto a common orientation whose x axis is the baseline, so epipolar lines become identical image rows and correspondence search collapses to 1D. We use the compact algorithm of Fusiello, Trucco and Verri [FTV00]: the new rotation has rows $(\mathbf{x}, \mathbf{y}, \mathbf{z})$ with $\mathbf{x}$ the unit baseline, $\mathbf{y} = \mathbf{k} \times \mathbf{x}$ ($\mathbf{k}$ the old optical axis of camera 1) and $\mathbf{z} = \mathbf{x} \times \mathbf{y}$; both views share the averaged intrinsics with zero skew, and the pixel maps are the homographies $\mathbf{H}_i = \mathbf{K}_{\text{new}}\mathbf{R}_{\text{new}}\mathbf{R}_i^\top\mathbf{K}_i^{-1}$. Since the rig is stationary, $\mathbf{H}_1, \mathbf{H}_2$ are computed once and each video frame is warped by table lookup.

On the rectified pair, dense correspondence is found by block matching: for each left pixel the sum of absolute differences (SAD) over a $(2w+1)^2$ window is minimized along the corresponding right row over a disparity range, followed by sub-pixel refinement by parabolic interpolation of the cost around the minimum. This is the classic local baseline method in the taxonomy of Scharstein and Szeliski [SS02] — chosen deliberately: it is simple, portable, trivially parallel, and its failure modes (textureless regions, occlusions, repetitive patterns) are well understood, giving a controlled baseline against which future matchers (rank/census transforms, semi-global aggregation) can be measured within the same harness.

Matched disparity d converts to metric depth by

$$Z = \frac{f\,B}{d}, \tag{7}$$

with f the shared rectified focal length (pixels) and B the baseline. Differentiating (7) gives the expected depth uncertainty

$$\sigma_Z \;=\; \frac{Z^2}{f\,B}\,\sigma_d, \tag{8}$$

for disparity noise σ_d — the quadratic growth with Z is the fundamental accuracy law of the rig and is verified experimentally in section 10.

7 From depth to models

Every depth estimate back-projects to a 3D point in the fixed world frame; points are accumulated in a growable cloud with optional per-point color and exported as ASCII PLY, directly loadable in MeshLab, CloudCompare or Blender for inspection. Over a video sequence the static-rig assumption makes accumulation a union in a common frame; per-point temporal statistics (persistence, variance) separate static structure from moving objects. The planned meshing path is classical: volumetric range-image fusion [CL96] or Poisson surface reconstruction [KBH06] over the accumulated cloud; both consume exactly the oriented-point data the pipeline produces, so they attach cleanly as later modules.

8 Planned extensions

8.1 Auxiliary photo collections

Still photographs of the same volume — arbitrary viewpoints, possibly a different camera — densify coverage and fill occlusions. The integration path follows the structure-from-motion recipe of Photo Tourism [SSS06]: detect scale-invariant features [Low04] in the photos *and* in the two calibrated video frames; match with a RANSAC gate [FB81]; register each photo to the rig's world frame by resection from matches against already-triangulated points; then re-triangulate all tracks with the registered views via the existing N-view DLT (section 5); finally refine everything with bundle adjustment [TMHF00]. The two calibrated cameras anchor the metric scale — the classic scale ambiguity of pure SfM does not arise. Dense multiview stereo over the registered set (e.g. patch-based methods [FP10]) is a further optional stage.

8.2 Range-finder fusion

A range finder (laser/ToF) contributes sparse but metrically strong depth. Two integration modes are planned: (i) if the range finder is rigidly mounted and extrinsically calibrated to the rig, each range sample is simply another world-frame point with small σ_Z, fused into the cloud with inverse-variance weights; (ii) if it produces an independent scan, the scan is registered to the vision cloud by closed-form absolute orientation [Hor87] initialized correspondence-free and refined with ICP [BM92]. In both modes the range data additionally serves as a ground-truth audit of the stereo depth law (8).

9 Numerical implementation notes

All decompositions reduce to one kernel: a one-sided (Hestenes) Jacobi SVD [GV13] for $m \geq n$ matrices, $n \leq 9$ in every present use (design matrices of (6) and the 8-point system). Jacobi SVD was chosen over Golub–Kahan bidiagonalization for its simplicity (∼100 lines, no workspace queries), unconditional convergence, and excellent relative accuracy on small matrices; its $O(mn^2)$-per-sweep cost is irrelevant at these sizes. Homogeneous solutions ($\mathbf{F}$, DLT points) are the right singular vector of the smallest singular value; rank-2 enforcement for $\mathbf{F}$ zeroes the smallest singular value and recomposes.

Conditioning is handled where it matters: the 8-point design matrix mixes terms of order 1, u, and u^2 ($u \sim 10^2$ pixels), giving condition numbers around 10^8 unnormalized; Hartley normalization [Har97] (centroid 0, RMS radius $\sqrt{2}$) reduces this to $O(10)$ and is applied always, with the result denormalized as $\mathbf{F} = \mathbf{T}_2^\top \hat{\mathbf{F}}\, \mathbf{T}_1$. All fundamental matrices are canonicalized (unit Frobenius norm, largest-magnitude entry positive) so that analytic and estimated matrices are directly comparable — this is what makes the drift monitor of section 4 a plain matrix norm.

The code is strict C99 (`-std=c99 -pedantic -Wall -Wextra` clean), depends only on `libm`, allocates only in unavoidable places (design matrices, images, clouds), and contains no hidden state except the demo's seeded generator. Determinism is treated as a feature: identical inputs give bit-identical outputs across runs, so every regression is reproducible.

10 Basic results

Two validation programs accompany the library. `tests/test_mv.c` checks exact properties on noiseless data (19 assertions: SVD reconstruction and orthonormality, inverse correctness, center recovery, ray/projection consistency, (4) to 10^{-8}, exact triangulation recovery, row alignment after rectification); all pass. `demo/synthetic.c` is the first controlled experiment: two cameras with $f = 800\,\text{px}$, 640×480 images, baseline $B = 0.5\,\text{m}$, the second camera yawed 6° inward, observing 250 points drawn uniformly in a $2 \times 1.5\,\text{m}$ window at depths 3.5–6 m; projections are corrupted with Gaussian noise $\sigma = 0.3\,\text{px}$ per coordinate (seed fixed). Table 2 summarizes the outcome.

Quantity	Value
points visible in both views	250/250
$\lVert \mathbf{F}_{\text{analytic}} - \mathbf{F}_{\text{8-point}} \rVert_F$ (unit-norm)	4.5×10^{-3}
mean symmetric epipolar distance (analytic $\mathbf{F}$)	0.345 px
mean symmetric epipolar distance (8-point $\mathbf{F}$)	0.341 px
triangulation RMS 3D error	0.0265 m
triangulation max 3D error	0.0876 m
RMS reprojection error	0.218 px
mean row misalignment $\lvert v_1 - v_2 \rvert$ before rectification	0.764 px
mean row misalignment after rectification	0.0 px

Table 2: Synthetic experiment, seed 42 (reproduce with `make demo`).

The numbers are consistent with theory. The epipolar residual $\approx 0.34\,\text{px}$ matches the injected noise level. The depth-error law (8) with $\bar{Z} \approx 4.7\,\text{m}$, $fB = 400\,\text{px}\cdot\text{m}$ and disparity noise $\sigma_d = \sqrt{2}\,\sigma \approx 0.42\,\text{px}$ predicts $\sigma_Z \approx 2.3\,\text{cm}$; the measured RMS 3D error of 2.65 cm (which also contains the smaller lateral components) agrees, confirming that the implementation performs at the noise floor of the geometry rather than being implementation-limited. The 8-point estimate reproduces the calibrated $\mathbf{F}$ to 4.5×10^{-3} in Frobenius norm from noisy data — tight enough to serve as the calibration-drift monitor of section 4. The reconstructed cloud is written to `out_cloud.ply` for visual inspection.

11 Roadmap

In dependency order: (1) feature detection/description and robust matching (enables live correspondences, RANSAC-gated by section 4); (2) image warping by the rectifying homographies with bilinear resampling, closing the dense video path end-to-end; (3) temporal cloud statistics for static/moving segmentation (section 7); (4) photo-collection registration (subsection 8.1); (5) Hartley–Sturm optimal triangulation and bundle adjustment; (6) range-finder fusion (subsection 8.2); (7) meshing. Each step lands as a new module behind the same conventions (Table 1) with its own noiseless-exact tests and a synthetic experiment extending Table 2.

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